Academic Medicine, 2026, 00, 1–9
https://doi.org/10.1093/acamed/wvag141
Advance access publication: April 28, 2026
Scholarly Perspective

ACADEMIC MEDICINE
A JOURNAL OF THE AAMC
OXFORD

Applying a learning analytics lifecycle framework to identify gaps in digital learning design: a worked example in health professions education

Zhenhua Xu, PhD[1], Jaimie Coleman, MSc[2], Amit Atrey, MD, MSc[3], Nhat Chau[4], Bryce Hunter[5], Ryan Brydges, PhD[1,6,*]

[1]Allan Waters Family Simulation Program, St. Michael's Hospital, Unity Health Toronto, Toronto, ON, Canada
[2]Department of Physical Therapy, Temerty Faculty of Medicine, University of Toronto, Toronto, ON, Canada
[3]Division of Orthopaedic Surgery, Department of Surgery, Temerty Faculty of Medicine, University of Toronto, Toronto, ON, Canada
[4]Dalla Lana School of Public Health, University of Toronto, Toronto, ON, Canada
[5]TRTN Inc, Toronto, ON, Canada
[6]Department of Medicine, Temerty Faculty of Medicine, University of Toronto, Toronto, ON, Canada
*Corresponding author: Ryan Brydges, Unity Health Toronto, 209 Victoria Street, Toronto, ON M5B 1X2, Canada. (ryan.brydges@utoronto.ca).

Abstract

Learning analytics involves collecting, analyzing, and visualizing the digital "footprints" that learners leave behind as they interact with digital learning environments. Learning analytics inform ongoing refinements to improve educational design and teaching practice. To date, research suggests that educators have not fully capitalized on learning analytics, despite the growing availability of data generated through digital learning platforms, programmatic assessment, and competency-based training. In this article, the authors adapted an established "learning analytics lifecycle" framework to evaluate a representative example of digital learning in HPE: *eduCAST,* an online, video-based website for orthopedic surgery training. The framework provided a stepwise approach for exploring the potential of learning analytics in *eduCAST*, identifying implementation gaps, and developing a transferable list of recommendations for educators in HPE. Demographic and engagement data of 141 registered *eduCAST* users were analyzed after being collected via website analytics. The findings revealed limitations and opportunities related to: (1) planning for the learning environment and its users, (2) the scope and specificity of available learning analytics data, (3) the purposeful use of data analysis techniques, and (4) the relative absence of educators' data-informed actions. Generic website analytics showed significant limitations: high website traffic did not correspond to meaningful learner engagement, nor did it provide useful data to inform ongoing website refinement. Indeed, these analytics produced more *data mysteries* than meaningful *data stories*. Consistent with prior reviews in the HPE literature, this worked example demonstrated that educators appear to make diminishing investments of effort and resources across the learning analytics lifecycle. The authors argue that HPE educators can benefit from using learning analytics frameworks to guide the design, implementation, evaluation, and long-term sustainability of digital learning environments. When thoughtfully collected, analyzed, and interpreted, learning analytics can enhance learners' experiences and outcomes, support educators' professional development, and inform continuous program refinement.

Keywords learning analytics framework, online learning, digital learning environments, instructional design, data analytics

Learning technologies, such as adaptive e-learning platforms, artificial intelligence agents, and video repositories, continue to reshape health professions education (HPE) by offering flexible and data-rich learning experiences.[1-3] As learners engage with such platforms, researchers and frontline educators have increasingly used learning analytics to collect, analyze, and interpret the digital "footprints" learners leave behind during their interactions.[4,5] By applying statistical and visualization techniques to analyze learning analytics data, educators can identify struggling learners, provide real-time feedback, and detect broader learner engagement patterns to guide improved educational designs, teaching practices, and institutional planning of educational resources.[4,6-12]

Despite the promises surrounding learning analytics, the HPE community remains in the early stages of adoption.[11,13] Early adopters have noted that the datasets from programmatic assessment, competency-based medical education (CBME),[10,14-16] and digital learning environments[17,18] present significant opportunities for collecting and integrating learning analytics, while also presenting ethical concerns.[19] For example, researchers have applied learning analytics to facilitate residents' professional development,[7] analyze medical students' success in online

Accepted: April 20, 2026

courses,[6,9] and explore the possibilities of "precision medical education."[16,20] However, we argue that most educators lack the time, training, or technical expertise to choose, implement, analyze, and act upon analytics effectively and ethically.[11,12] Consequently, educators who design digital learning environments appear to rely mostly on generic website analytics (eg, Google Analytics) that produce some engagement data (eg, page views, click sequences, session durations), yet do not regularly appear to inform adaptive improvements to their educational designs.[15,21-23]

In HPE, educators opting to collect data simply because they are accessible (eg, through available web analytics), or without a clear set of educational intentions, can produce "data waste."[24] Scholars in the business and computing systems literature refer to "dark data" as data that have been collected, and potentially processed and maintained in a system, yet somehow are under-utilized or un-analyzed, and thus remains "invisible" to decision-makers.[25,26] Such data may be holding untapped value, or they may represent data that do not add value and thus are merely "taking up space."[27] In academic research, for example, researchers estimate that unused data across research-focused universities represents a financial loss of ~7% of their research and development budget.[28] In the education setting, we interpret "data waste" as implying the collection of learner and instructional data that are either not meaningfully analyzed, misaligned with educational goals, or provide low signal relative to the effort required for collection, which ultimately results in minimal or no educational benefit. Data waste can be dually burdensome: it can overwhelm educators who lack the tools or capacity to extract meaningful insights, and it can fatigue learners who certainly notice the increased collection of their data alongside little to no corresponding educational benefit. In this early stage of adoption in HPE, researchers and educators need to identify comprehensive frameworks that can guide their effective selection and use of learning analytics, to avoid creating "dark data" or "data graveyards" of unused, yet useful data. Established frameworks also help to anticipate challenges and choices that may inadvertently lead to data waste.

Frameworks for implementing learning analytics offer educators a structured approach to anticipate and address previously identified challenges.[29-33] Over the past decade, education scientists have collected substantial evidence to propose a comprehensive learning analytics life cycle framework with four iterative stages (Figure 1).[13,30,33] The life cycle begins with educators planning for which digital footprints (eg, click sequences, time spent on different pages) to collect, including how to collect them and for what purposes. Once collected, the data then need to be processed and analyzed effectively to generate meaningful interpretations. Finally, educators must act on the data to better

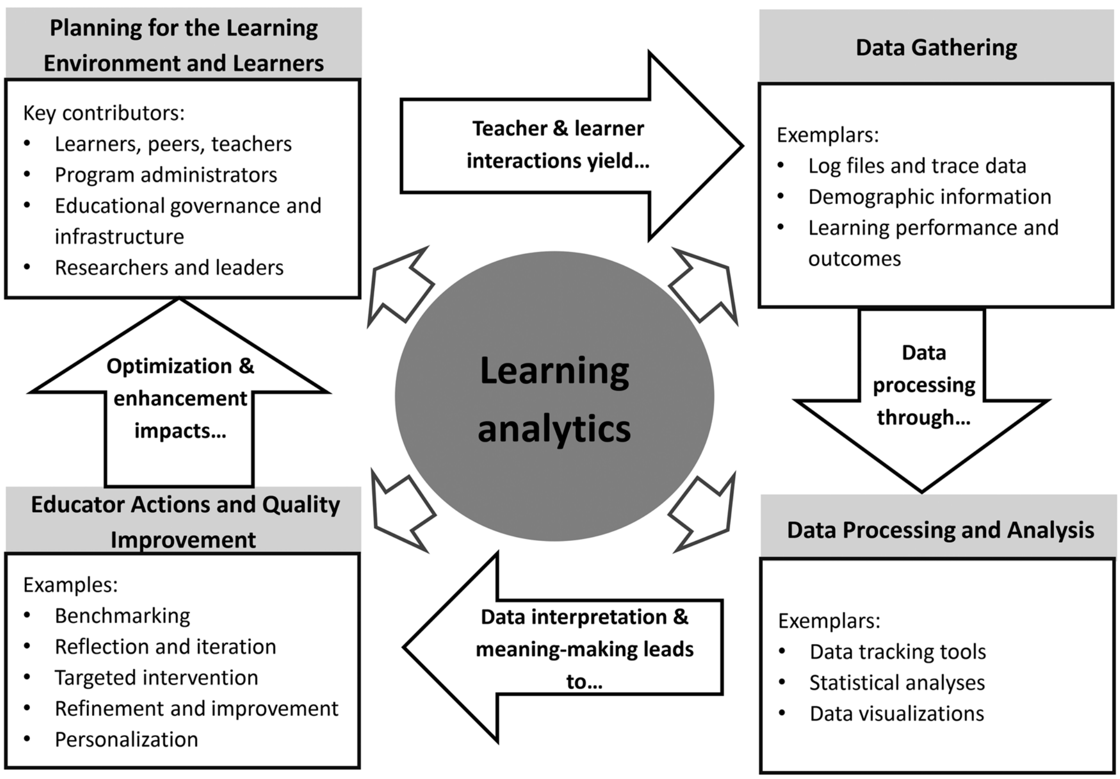

Figure 1 Four-stage learning analytic lifecycle framework adapted with permission from Khalil M, Ebner M. Learning Analytics: Principles and Constraints. *Learning Analytics*. 2015.[33] It illustrates the use of learning analytics data to support continuous improvement in digital learning environments. The cycle begins with the learning environment and the key people, including learners, teachers, program administrators, and broader educational infrastructure. Their interactions with the learning environment generate *datasets*, such as log files, digital trace data, demographic information, and measures of learning performance. These data are *processed and analyzed* using established techniques, including data tracking tools, statistical analyses, and data visualizations. The findings derived from these analyses can inform *educators' actions and quality improvement* strategies, such as benchmarking, ongoing refinement of plans, targeted instructional interventions, and assessment decisions. These actions subsequently enhance and reshape the learning environment, creating an ongoing cycle of evidence-informed quality improvement.

understand and support learners, refine the program, and improve their own teaching approaches. A recent review found that most HPE studies focus on early stages (ie, planning and data gathering), with few examples of effective approaches for the latter two stages (ie, analyzing and data-informed actions).[13,15,17,21] This gap demonstrates that while educators in HPE have initiated the process of generating data, they appear to be struggling to analyze, interpret, and make meaningful changes with that data.[11,13]

In this article, we argue that applying a learning analytics life cycle framework (from here referred to as "the framework"), even to a resource-constrained digital learning environment, can yield valuable lessons for educators in HPE. We applied the framework to *eduCAST*, an open-access, video-based platform for orthopedic surgery education, as a worked example of what we consider a "typical" HPE educator's approach to identifying their teaching passion, creating a digital learning environment to address perceived gaps and needs in a learner population, and collecting available analytics with good intentions for their future use. By applying the life cycle framework to evaluate the *eduCAST* platform, we aimed to demonstrate how to diagnose implementation gaps, identify promising and context-specific analytics, and generate a list of actions for adapting future educational designs. Using a structured framework to generate such data and associated recommendations, we provide a guide that equips educators, in HPE and other education domains, with a systematic approach to evaluate their own educational design plans. Ultimately, we aimed to provide a practical starting point for educators seeking to learn more about learning analytics and to feel informed enough to consider integrating them into their educational design and evaluation practices.

Scholarly approach

Selecting a framework

Clow's[30] original learning analytics framework has been used and refined by other researchers in the learning sciences to guide empirical evaluations of how programs and organizations use learning analytics.[13,29,30,33] In particular, we judged Khalil and Ebner's[33] framework as the most evidence-based, well-defined, and detailed version, which has also been cited heavily in the learning sciences. With permission, we made small adjustments to their four-stage framework, representing it as: (1) *planning for the learning environment and learners*, which characterizes intentionally considering and designing analytics for the instructional context and intended user groups, (2) *gathering data* generated by the learning environment and its users, capturing various aspects of learner and educator activity, (3) *data processing and analysis,* including the techniques for processing and analyzing raw data into interpretable signals, and (4) *educator actions and quality improvement*, where data-driven and objective-informed decisions are implemented to refine learning, and/or program goals. Figure 1 visualizes the framework as a foundational guide to help educators in identifying key users, informing data sampling and analysis, and interpreting outcomes to support iterative refinements of instructional design.

Educational context

In 2022, a surgeon and educator (author AA) collaborated with an educational technologist (author BH) to design *eduCAST* as an open-access repository of physician-generated and approved videos. They aimed to create a more dynamic way to teach the complexities of orthopedic surgery knowledge, and to promote equity in access amongst trainees globally (ie, many jurisdictions do not have access to high-quality multimedia resources). The videos (12-20 mins each) incorporate multimedia elements (eg, 3D animations, analogies, anecdotes, and clinical footage) to explain complex orthopedic concepts. They organized content into six thematic areas, such as hips, knees, and shoulders, each containing 5-12 topic-specific videos (eg, hip biomechanics, surgical approaches to hip arthroplasty). Any user can access the entire website by registering for a free account. The intended learners for *eduCAST* include medical students, postgraduate trainees, and other health professionals seeking to advance their knowledge and skills in orthopedic surgery. While widely accessible, *eduCAST* functions as an extra-curricular resource and is not formally embedded in any medical school or residency program curricula. Figure 2 provides an overview of *eduCAST's* components.

Project context

In early 2023, our team began collaborating on how to understand the impact and value of *eduCAST* for its learners. Given the team's resource limitations, we learned quickly that website analytics represented the only ongoing and available data they had recorded about their learners. In reviewing the literature,[21-23] we understood the *eduCAST* team experienced a common scenario: open-access, video-based content developed under resource constraints. We judged that these characteristics made *eduCAST* ideal for testing the framework's use as a gap diagnosis and analysis guide. We applied the framework to generate foundational knowledge about *eduCAST's* challenges and opportunities, with a plan to integrate context-specific and custom learning analytics into the system as part of an ongoing program of research.

Applying the learning analytics life cycle framework

To operationalize the framework, we first identified key people based on their direct involvement in the design, implementation, and use of the learning platform. This included authors AA and BH, respectively, as the educational lead who produced and curated the educational videos, and the educational technologist who shaped the learning platform's architecture. We also considered learners who had left "digital footprints" when accessing *eduCAST*. Our team collectively discussed, operationalized, and conducted custom analyses relating to each stage of the life cycle. In our attempts to adapt and apply the framework to academic medicine, we aimed to reflect real-world conditions and show how educators can operationalize the framework even when data are limited.

Figure 2 Examples of eduCAST website features, including (Panel A) landing page organized by specialty content, (Panel B) screenshots from hip video chapters, and (Panel C) visuals from 3D anatomical learning module.

Planning for the learning environment and learners in *eduCAST*

A pivotal initial step in the life cycle involves identifying and understanding context and users. The *eduCAST* team leveraged the built-in analytics service tools (ie, web analytics and Google Analytics) to collect user demographic information, including IP addresses and locations, which enabled tracking of returning users and general engagement trends.

Gathering data in *eduCAST*

Our next step involved reviewing and evaluating the system-generated usage data. As suggested in the educational data mining literature, we focused on timing (eg, time spent) and frequency (eg, log-ins, videos viewed) indicators of learner engagement that reflect persistence, depth of interaction, and breadth of content exposure.[6,9,18] We selected these variables because they were the only system-generated metrics consistently captured by *eduCAST* and because they enabled us to examine interaction patterns relevant to our diagnostic goal of determining users' engagement patterns. In June 2023, we prompted the *eduCAST* team to create a data archive, which they had not been doing historically. That involved saving and storing the available usage data, such as the number of log-ins, time spent on various parts of the website, and videos viewed. We then chose a convenience sampling period, from June 2023 to October 2023, during which author BH stored and pooled all user data for our analysis.

Data processing and analysis in *eduCAST*

We aimed to illustrate how educators in HPE might begin to explore and analyze available generic website analytics data (or similar) associated with their digital learning environments. After receiving the available *eduCAST* user data, four authors (ZX, NC, RB, JC) met regularly to formulate an analysis plan. Our plan was formed around clarifying the nature of the website analytics data (eg, page views, session duration on videos) and how they had been organized for each video included on the platform.

We developed a custom coding framework deductively using literature on technology-enhanced learning, principles of multimedia learning, and research on massive open online courses.[34-36] We piloted the application of our coding framework to evaluate two high-access and two low-access videos on the website,[37] with the aim of understanding patterns in how *eduCAST* users interacted with these videos (ie, could we generate information about what made videos more or less engaging?). Three coders (ZX, JC, NC) applied the framework to document video content (11 items), video adherence to multimedia learning principles (4 items), instructional media dimensions (10 items), and user interface (6 items). We calculated interrater reliability using Cohen's Kappa. Given the limitations of the available data, we conducted descriptive analyses of the available time data for each selected video (ie, means and standard deviations).

Educator actions and quality improvement in *eduCAST*

We probed whether and how the *eduCAST* team used data sources and available analytics to inform refinements to the platform. We documented these actions through our email exchanges and meeting discussions. We applied a simple content analysis approach to summarize and synthesize these data sources.[38]

Team reflexivity in applying the framework

Our team members' diverse perspectives certainly influenced how we designed and interpreted our worked example. As an educational psychologist, ZX selected and led in applying the learning analytics framework. As an education scientist with expertise in self-regulated learning and HPE research, RB believes strongly that worked examples using meaningful frameworks can enhance educators' design and delivery of digital learning. As a physiotherapy educator and PhD student in HPE research, JC has learned to develop many custom digital learning environments. Similarly, as a student in HPE, NC has experienced digital learning environments that vary greatly in how they help him learn. As the *eduCAST* designers wish to demonstrate their impact, AA is an orthopedic surgeon and clinical educator who supervises and teaches surgical trainees, and who produces and curates the educational videos on *eduCAST*. BH has expertise in medical media technology and expressed a strong desire to design more meaningful analytics into the *eduCAST* platform. Collectively, we adopted a constructivist lens by constantly engaging in reflexive dialogue to explore how we each personally experienced the processes of evaluating *eduCAST*, and how our situated perspectives continually shaped and reshaped how we applied the framework.

Findings from the worked example

Below, we report on all four stages of the life cycle (Figure 3), to illustrate how we populated the empty cells in Figure 1 with our findings. Notably, we present the findings as a worked example to demonstrate the value educators might gain by applying the framework to any digital learning environment, rather than as a critique of *eduCAST* specifically. Our findings demonstrate how using the framework diagnosed implementation gaps, identified

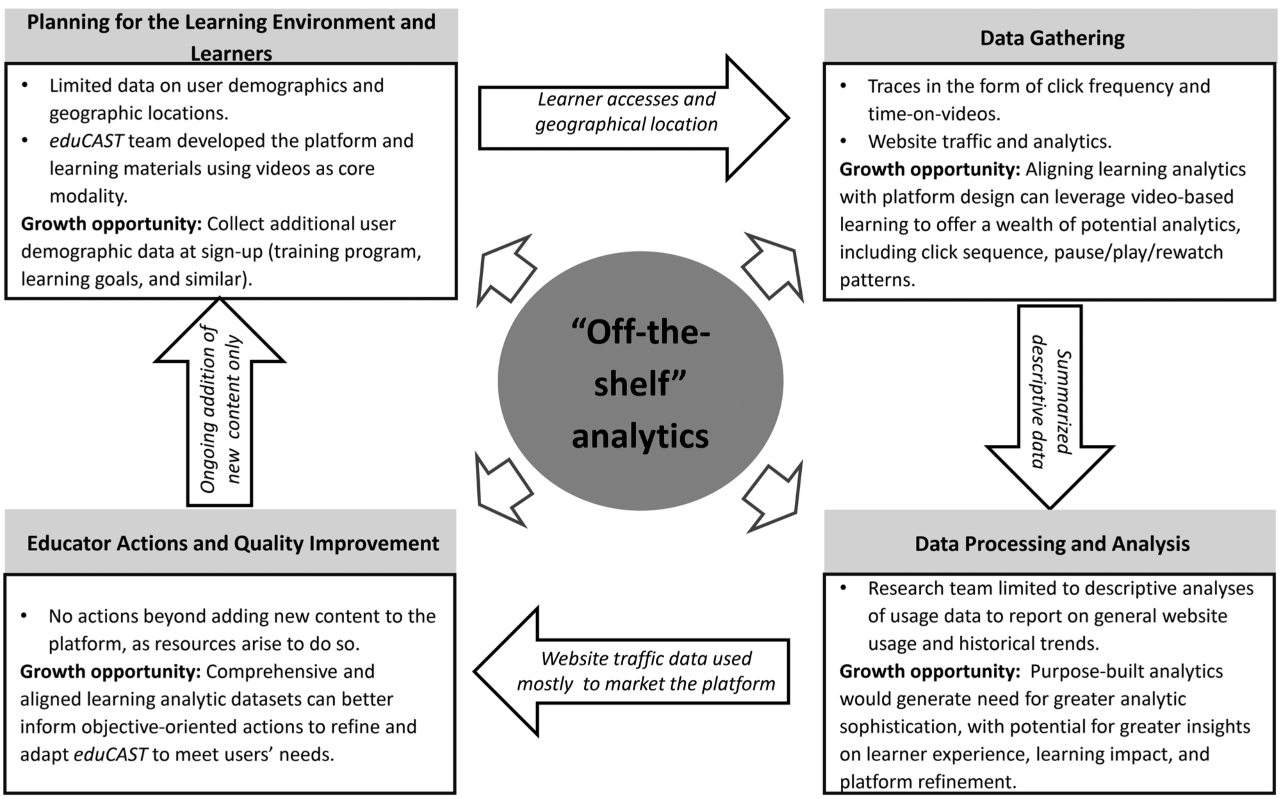

Figure 3 Mapping of our study findings onto the learning analytics life cycle framework. Boxes highlight challenges and growth opportunities across learners and environments, datasets, analytic techniques, and objective-informed actions.

promising learning analytics, and generated a list of actions for adapting future designs of the platform. We present our findings and a summary of insights from each stage of the framework below.

Stage 1: Identifying collected learner and contextual data in *eduCAST*

Our project plan yielded a sample of 141 registered *eduCAST* users with available system-generated usage metrics, a relatively robust cohort that reflected the active user population during the June-October 2023 study window. The data included users from 24 countries, with most from Canada (29.1%) and the United States of America (27.7%). Beyond basic user geolocation data, *eduCAST* did not collect demographic information. Thus, we could not discern any other meaningful information to identify unique learner characteristics. For instance, we could not conclude whether learners accessing *eduCAST* were orthopedic trainees accessing it for training purposes, members of the general public accessing it to reduce their concerns about an upcoming surgery, or any other possible groups with a related interest. Hence, we identified an implementation gap, suggesting educators in HPE would benefit from explicitly collecting de-identified demographic data from all users to facilitate unique learner-driven improvements.

Stage 2: Gathering available data in *eduCAST*

We found that *eduCAST's* chosen analytics generated datasets on website traffic that were limited in scope and depth, such as the click frequency and average visit duration for specific learning materials (eg, chapters, videos).

Stage 3: Seeking or adapting data processing and analysis techniques in *eduCAST*

The *eduCAST* team used *descriptive analytics* to visualize website traffic and identify trends. For example, the data in Table 1 show that users visited the "Biomechanics of the hip" video far more often, though spent less time per visit than the "Hallux Valgus correction" video, suggesting differential engagement. However, without additional data, we could not determine whether users' engagement patterns with different videos were related to differences in instructional design, clinical relevance, user preferences, or other factors. Table 1 demonstrates that these descriptive characteristics and engagement data yield limited insights regarding clear patterns that educators might capitalize on—it does seem, for instance, that shorter videos were associated with more visits and longer time spent per visit. Hence, our findings suggest that generic website analytics data provide limited utility as educational tools, and that HPE educators would benefit from brainstorming platform-specific analytics that could inform instructional refinement and/or assess learning impact.

When piloting our coding framework on the four selected videos, the three coders achieved acceptable inter-rater reliability with Cohen's kappa values ranging from 0.61 to 0.80, except for the video style category (poor agreement). We found that all videos utilized the key principles and guidelines of multimedia learning across the three dimensions (ie, video style, video content, and user interface). We observed frequent use of multimedia learning principles (eg, focusing on the instructor's body and hand, 2/3D animation, live action) in all videos. Low-access videos did not demonstrate learning objectives; whereas high-access videos included signaling cues more often and were shorter in duration compared to low-access videos. While limited, these findings suggest some characteristics that educators can attend to when creating video content, including the use of multimedia learning principles in designing the content and attending to video length.[37]

Stage 4: Seeking evidence of educator actions and quality improvement in *eduCAST*

Our analysis of team discussions showed that, historically, the *eduCAST* team leveraged website analytics data to showcase website usage, to secure funding, or to promote partnerships for content expansion. Beyond that, the team had not translated the website analytics data into actionable educational instructional

Table 1 Illustrative example of videos with variable engagement according to *eduCAST* usage metrics collected from June to October 2023.

Chapter names	Mean visit frequency (per month)	Video length (mm:ss)	Mean time per visit (mm:ss)
Chapter 1: Biomechanics of the hip	2461	11:12	6:58
Chapter 1: Anatomy and biomechanics of the knee	1516	15:52	8:03
Chapter 2: History of hip arthroplasty	1436	20:24	8:17
Chapter 3: Total shoulder arthroplasty	153	17:48	10:31
Chapter 4: ACL reconstruction	100	29:35	4:38
Chapter 2: Hallux valgus correction	77	23:56	12:16
Chapter 6: Complications of shoulder replacement and treatment	55	14:00	9:46
Chapter 4: Ankle arthritis	39	16:36	9:23
Chapter 5: Total ankle arthroplasty	38	27:37	2:33

This table summarizes the most frequently accessed videos, based on system-generated analytics from 141 registered *eduCAST* users. These examples illustrate how basic usage metrics can inform preliminary insights into engagement patterns relevant to instructional and/or content refinement.

strategies and meaningful interventions. On reflection, the *eduCAST* team reported a disconnect in their priorities: by emphasizing usage data alone, they had missed any opportunity to generate and use learning analytics as a mechanism for continuous platform educational improvement.

Lessons learned and recommendations

We believe that the challenges and disconnects we encountered in applying the framework to *eduCAST* are not unique. Indeed, the *eduCAST* team developed the website with a finite budget, relied on built-in website analytics (eg, Google Analytics), collected minimal demographic and user motivation and engagement data, and did not regularly analyze and interpret data to inform improvements to the educational design. These results align with a recent review of HPE literature,[13] showing diminishing investments from educators into all stages, and especially the latter stages of the life cycle.

Applied to *eduCAST*, our recommendations (Figure 3) include to: (1) collect more comprehensive demographics during the user registration process (based on stage 1 findings), (2) ask users to provide data on their prior orthopedic knowledge, and their reasons for accessing the website (based on stage 1 findings), (3) analyze the demographic and prior knowledge data to inform the selection of purpose-built analytics that capture interactions with the primarily video-based content (based on stage 2 and stage 3 findings), and (4) consider introducing assessments, such as knowledge tests, to allow analyses of how users' data relate to their learning outcomes (based on stage 4 findings). In the sections below, we connect these findings to generalizable principles and provide recommendations for HPE educators to consider as they design and refine their digital learning environments.

Our findings for *eduCAST*'s design and use of learning analytics reflect how *implementation gaps* arise when educators in HPE develop and deploy digital learning environments without simultaneously building the necessary data literacy to select and design analytics that inform educational decision-making.[16,21,38,39] Without a clear plan or framework to guide them, educators may end up choosing analytics that offer little information on how and why learners engage with content to achieve specific learning outcomes. For example, we struggled to interpret users' usage patterns for the *eduCAST* videos: what factors contributed to higher or lower video view counts? With *eduCAST*, these potential "data stories" ultimately became data mysteries as we found ourselves limited to descriptive analyses.

Generally, educators report receiving limited organizational support to learn how to develop learning analytics, implement them into their practice, and access, analyze, and interpret the resulting data.[39,40] As Clow[30] noted: "analytics cannot achieve their potential without practitioners embedding them into pedagogic contexts." Our findings suggest that educators can *generate a list of preliminary and ongoing actions* that strengthen their capacity to manage all stages of the life cycle. In the planning and early implementation phases, we recommend that educators meaningfully involve learners from diverse groups to ensure their learning needs and objectives inform the selection and integration of analytics into the digital learning environment.[40,41] Next, educators who lack the time and expertise to analyze and interpret data may benefit from developing collaborations to support how they interpret, act on, and use the data to refine their learning and teaching processes.[42-44] Applying these recommendations to *eduCAST*, we suggest asking learners to respond to structured needs assessments during website registration, or engaging trainees from multiple Departments of Orthopaedics, which could clarify the demographic data learners will share comfortably with the site, their intentions for using the platform, their needs as they navigate the content, and the forms of feedback and data they value.

Our findings also revealed how misaligned analytics can stall educators' attempts to engage in quality improvement, and implicated some *promising and context-specific analytics* for *eduCAST*'s designers to pursue. When educators use their instructional design (ie, learning activities, available resources, and assessments) as the frame of reference[45,46] for understanding which learning analytics data to collect, for what purposes, and how to make meaning from them, then the value of each data source increases.[29] With its primarily video-based content, for instance, we recommend that *eduCAST*'s designers develop learning analytics that trace users' interactions with the videos (eg, the sequences in which they watch them, how they pause, stop, and restart segments, and so on), which researchers have shown can yield more diagnostic data for ongoing refinement of video-based content.[17]

Conclusions

To amplify the translational value of our worked example of *eduCAST*, we proposed the following approach that HPE educators can adopt when considering whether and how to integrate learning analytics into their educational designs. First, like most educational activities (eg, research, validation, evaluation), educators should select a structured framework to guide their design, delivery, and ongoing refinements to their digital learning environments. Second, educators must invest effort into all four stages of the learning analytics life cycle to create alignment and impact through an interconnected approach to designing digital learning environments and associated analytics. Third, educators must seek broad input from learners, colleagues, and program leaders to determine how best to capture details about why learners access their digital learning platforms and for what purposes, to guide the selection of meaningful, actionable, and learner-centered learning analytics. When thoughtfully collected, analyzed, and interpreted, learning analytics data can drive meaningful improvements in learners' experiences and outcomes, support educators' professional development, and inform continuous program refinement.

Conflicts of interest

A.A. and B.H. are developers of *eduCAST*, though neither profit from its ongoing use or growth.

Funding

R.B. is thankful to Anne and David Patterson for generously supporting the Chair in Advancing Technology-Enhanced Learning,

held at St. Michael's Hospital, Unity Health Toronto & University of Toronto.

Ethical approval

This study was approved by the Research Ethics Board at the University of Toronto (Protocol No. 00044696) and the Research Ethics Board at St. Michael's Hospital, Unity Health Toronto (REB No. 20-239).

Disclaimers

None declared.

Previous presentations

Some content of this manuscript was previously presented at the American Educational Research Association Annual Meeting, April 11, 2024, Philadelphia, Pennsylvania. The authors cite the associated online paper in the References section.

Data availability

The data underlying this article will be shared on reasonable request to the corresponding author.

References

1. MacNeill H, Masters K, Nemethy K, Correia R. Online learning in Health Professions Education. Part 1: teaching and learning in online environments: AMEE Guide No. 161. *Med Teach*. 2024;46:4-17. https://doi.org/10.1080/0142159X.2023.2197135
2. Gordon M, Daniel M, Ajiboye A, et al. A scoping review of artificial intelligence in medical education: BEME Guide No. 84. *Med Teach*. 2024;46:446-470. https://doi.org/10.1080/0142159X.2024.2314198
3. Niekrenz L, Spreckelsen C. How to design effective educational videos for teaching evidence-based medicine to undergraduate learners—systematic review with complementing qualitative research to develop a practicable guide. *Med Educ Online*. 2024;29:2339569. https://doi.org/10.1080/10872981.2024.2339569
4. Banihashem SK, Noroozi O, Van Ginkel S, Macfadyen LP, Biemans HJA. A systematic review of the role of learning analytics in enhancing feedback practices in higher education. *Educ Res Rev*. 2022;37:100489. https://doi.org/10.1016/j.edurev.2022.100489
5. Siemens G. Learning analytics: the emergence of a discipline. *Am Behav Sci*. 2013;57:1380-1400. https://doi.org/10.1177/0002764213498851
6. Saqr M, Fors U, Tedre M. How learning analytics can early predict under-achieving students in a blended medical education course. *Med Teach*. 2017;39:757-767. https://doi.org/10.1080/0142159X.2017.1309376
7. Holmboe ES, Yamazaki K, Nasca TJ, Hamstra SJ. Using longitudinal milestones data and learning analytics to facilitate the professional development of residents: early lessons from three specialties. *AcadMed*. 2020;95:97-103. https://doi.org/10.1097/ACM.0000000000002899
8. Matcha W, Gašević D, Uzir NA, Jovanović J, Pardo A. Analytics of learning strategies: associations with academic performance and feedback. In: *Proceedings of the 9th International Conference on Learning Analytics & Knowledge*. ACM; 2019:461-470. https://doi.org/10.1145/3303772.3303787
9. Cirigliano MM, Guthrie CD, Pusic MV. Click-level learning analytics in an online medical education learning platform. *Teach Learn Med*. 2020;32:410-421. https://doi.org/10.1080/10401334.2020.1754216
10. Carney PA, Sebok-Syer SS, Pusic MV, Gillespie CC, Westervelt M, Goldhamer MEJ. Using learning analytics in clinical competency committees: increasing the impact of competency-based medical education. *Med Educ Online*. 2023;28:2178913. https://doi.org/10.1080/10872981.2023.2178913
11. Thoma B, Warm E, Hamstra SJ, et al. Next steps in the implementation of learning analytics in medical education: consensus from an International Cohort of Medical Educators. *J Grad Med Educ*. 2020;12:303-311. https://doi.org/10.4300/JGME-D-19-00493.1
12. Ten Cate O, Dahdal S, Lambert T, et al. Ten caveats of learning analytics in health professions education: a consumer's perspective. *Med Teach*. 2020;42:673-678. https://doi.org/10.1080/0142159X.2020.1733505
13. Bojic I, Mammadova M, Ang CS, et al. Empowering health care education through learning analytics: in-depth scoping review. *J Med Internet Res*. 2023;25:e41671. https://doi.org/10.2196/41671
14. Chan T, Sebok-Syer S, Thoma B, Wise A, Sherbino J, Pusic M. Learning analytics in medical education assessment: the past, the present, and the future. Promes S, ed. *AEM Educ Train*. 2018;2:178-187. https://doi.org/10.1002/aet2.10087
15. Chan AK, Botelho MG, Lam OL. Use of learning analytics data in health care-related educational disciplines: systematic review. *J Med Internet Res*. 2019;21:e11241. https://doi.org/10.2196/11241
16. Thoma B, Ellaway RH, Chan TM. From utopia through dystopia: charting a course for learning analytics in competency-based medical education. *AcadMed*. 2021;96:S89-S95. https://doi.org/10.1097/ACM.0000000000004092
17. Yoon M, Lee J, Jo IH. Video learning analytics: investigating behavioral patterns and learner clusters in video-based online learning. *Internet High Educ*. 2021;50:100806. https://doi.org/10.1016/j.iheduc.2021.100806
18. Xu Z, Zdravkovic A, Moreno M, Woodruff E. Understanding optimal problem-solving in a digital game: the interplay of learner attributes and learning behavior. *Comput Educ Open*. 2022;3:100117. https://doi.org/10.1016/j.caeo.2022.100117
19. Marcotte K, Yang P, Millis MA, et al. Ethical considerations of using learning analytics in medical education: a critical review. *Adv Health Sci Educ*. 2025;30:87-101. https://doi.org/10.1007/s10459-024-10402-7
20. Desai SV, Burk-Rafel J, Lomis KD, et al. Precision education: the future of lifelong learning in medicine. *AcadMed*. 2024;99:S14-S20. https://doi.org/10.1097/ACM.0000000000005601
21. Wilson A, Watson C, Thompson TL, Drew V, Doyle S. Learning analytics: challenges and limitations. *Teach Higher Educ*.

2017;22:991-1007. https://doi.org/10.1080/13562517.2017.1332026
22. Fernandez-Nieto GM, Echeverria V, Shum SB, et al. Storytelling with learner data: guiding student reflection on multimodal team data. *IEEE Trans Learning Technol*. 2021;14:695-708. https://doi.org/10.1109/TLT.2021.3131842
23. Alby T. Popular, but hardly used: has Google Analytics been to the detriment of Web Analytics? In: *Proceedings of the 15th ACM Web Science Conference 2023*. ACM; 2023:304-311. https://doi.org/10.1145/3578503.3583601
24. Rahmouni M. Dark data: why what you don't know matters: by David J. Hand, Princeton, NJ: Princeton University Press, 2020, xii + 330 pp., ISBN 9780691182377. *Technometrics*. 2023;65:129-131. https://doi.org/10.1080/00401706.2022.2163804
25. Marumolwa L, Marnewick CMC. Unveiling dark data in organisations—sources, challenges, and mitigation strategies. *Int J Serv Sci Manag Eng Technol*. 2025;16:1-32. https://doi.org/%2010.4018/IJSSMET.386167
26. Ravindranathan P, Ashok P, Prabhu S. Illuminating the dark: gaining insights and managing risks with dark analytics. In: *2024 International Conference on Intelligent and Innovative Technologies in Computing, Electrical and Electronics (IITCEE)*. IEEE; 2024:1-6. https://doi.org/10.1109/IITCEE59897.2024.10467380
27. Hasan R, Burns R. The Life and Death of Unwanted Bits: Towards Proactive Waste Data Management in Digital Ecosystems. In: *Third International Conference on Innovative Computing Technology (INTECH 2013)*; 2013:144-148. https://doi.org/10.48550/arXiv.1106.6062
28. Bowers EC, Stephenson J, Furlong M, Ramos KS. Scope and financial impact of unpublished data and unused samples among U.S. academic and government researchers. *iScience*. 2023;26:107166. https://doi.org/10.1016/j.isci.2023.107166
29. Wise AF, Vytasek J. Learning analytics implementation design. In: Lang C, Siemens G, Wise A, Gasevic D, eds. *Handbook of Learning Analytics*. 1st ed. Society for Learning Analytics Research (SoLAR); 2017:151-160. https://doi.org/10.18608/hla17.013
30. Clow D. The learning analytics cycle: closing the loop effectively. In: *Proceedings of the 2nd International Conference on Learning Analytics and Knowledge*. ACM; 2012:134-138. https://doi.org/10.1145/2330601.2330636
31. Syed M, Anggara T, Lanski A, Duan X, Ambrose GA, Chawla NV. Integrated closed-loop learning analytics scheme in a first year experience course. In: *Proceedings of the 9th International Conference on Learning Analytics & Knowledge*. ACM; 2019:521-530. https://doi.org/10.1145/3303772.3303803
32. Rodríguez-Triana MJ, Prieto LP, Martínez-Monés A, Asensio-Pérez JI, Dimitriadis Y. The teacher in the loop: customizing multimodal learning analytics for blended learning. In: *Proceedings of the 8th International Conference on Learning Analytics and Knowledge*. ACM; 2018:417-426. https://doi.org/10.1145/3170358.3170364
33. Khalil M, Ebner M. *Learning Analytics: Principles and Constraints*; 2015.
34. Chorianopoulos K. *A Taxonomy of Video Lecture Styles*. IRRODL. 2018. https://doi.org/10.19173/irrodl.v19i1.2920
35. Mayer RE. ed. *The Cambridge Handbook of Multimedia Learning*. 2nd ed. Cambridge University Press; 2014.
36. Cook DA, Ellaway RH. Evaluating technology-enhanced learning: a comprehensive framework. *Med Teach*. 2015;37:961-970. https://doi.org/10.3109/0142159X.2015.1009024
37. Xu Z. Exploring the educational affordance of an open access video-based orthopedic surgery learning portal. In: *Proceedings of the 2024 AERA Annual Meeting*. AERA; 2024. https://doi.org/10.3102/2107786
38. Alzahrani AS, Tsai YS, Iqbal S, et al. Untangling connections between challenges in the adoption of learning analytics in higher education. *Educ Inf Technol (Dordr)*. 2023;28:4563-4595. https://doi.org/10.1007/s10639-022-11323-x
39. Shibani A, Knight S, Buckingham Shum S. Educator perspectives on learning analytics in classroom practice. *Internet High Educ*. 2020;46:100730. https://doi.org/10.1016/j.iheduc.2020.100730
40. Gray G, Schalk AE, Cooke G, Murnion P, Rooney P, O'Rourke KC. Stakeholders' insights on learning analytics: perspectives of students and staff. *Comput Educ*. 2022;187:104550. https://doi.org/10.1016/j.compedu.2022.104550
41. Wise AF. Learning analytics: using data-informed decision-making to improve teaching and learning In: Adesope OO, Rud AG, eds. *Contemporary Technologies in Education*. Springer International Publishing; 2019:119-143. https://doi.org/10.1007/978-3-319-89680-9_7
42. Dawson S, Pardo A, Salehian Kia F, Panadero E. An integrated model of feedback and assessment: from fine grained to holistic programmatic review. In: *LAK23: 13th International Learning Analytics and Knowledge Conference*. ACM; 2023:579-584. https://doi.org/10.1145/3576050.3576074
43. Mahmoud M, Dafoulas G, Abd ElAziz R, Saleeb N. Learning analytics stakeholders' expectations in higher education institutions: a literature review. *IJILT*. 2020;38:33-48. https://doi.org/10.1108/IJILT-05-2020-0081
44. Wollny S, Di Mitri D, Jivet I, et al. Students' expectations of learning analytics across Europe. *Comput Assist Learn*. 2023;39:1325-1338. https://doi.org/10.1111/jcal.12802
45. Macfadyen LP, Lockyer L, Rienties B. Learning design and learning analytics: snapshot 2020. *JLA*. 2020;7:6-12. https://doi.org/10.18608/jla.2020.73.2
46. Lim L, Bannert M, Van Der Graaf J, et al. Effects of real-time analytics-based personalized scaffolds on students' self-regulated learning. *Comput Human Behav*. 2023;139:107547. https://doi.org/10.1016/j.chb.2022.107547